

Macroeconomics

Ingyu Bahng

June 2026

Contents

1	Fundamental Concepts	2
2	Economic Indicators	5
2.1	Gross Domestic Product	6
2.2	Labor Market	8
2.3	Inflation	10
2.4	Housing	12
2.5	Sentiment Indicators	13
2.6	Economic Calendar and Indicator Releases	13
3	Macroeconomic Relationships	14
3.1	Aggregate Demand	14
3.2	Aggregate Supply	16
3.3	Economic Theory	17
4	Financial Sector	22
4.1	Money and the Money Market	22
4.2	Banking System	24
5	Stabilization Policies	27
5.1	Fiscal Policy	27
5.2	Monetary Policy	28
6	International Trade	30
6.1	Balance of Payments	30
6.2	Exchange Rates	32

1 Fundamental Concepts

Economics is the social science that studies the production, distribution, and consumption of goods and services. At its core, economics addresses the fundamental problem of **scarcity**: while resources such as labor, capital, and natural resources are limited, human wants are essentially unlimited. This creates the central challenge that all economies must navigate—how to allocate scarce resources among competing uses.

Macroeconomics is the branch of economics that studies the economy as a whole, including phenomena such as economic growth, inflation, unemployment, and national income. Unlike microeconomics, which focuses on individual behavior, macroeconomics examines aggregate outcomes and the policies that influence them.

Core Concept 1.1: Factors of Production

The production of goods and services requires four essential factors of production:

- (a) **Land** encompasses all natural inputs used in production, including agricultural land, mineral deposits, water resources, and the physical space upon which production facilities are constructed.
- (b) **Labor** represents the human effort devoted to production, encompassing both physical and mental work. Labor includes the services of workers across all sectors, from agricultural laborers harvesting crops to knowledge workers developing software.
- (c) **Capital** refers to goods that are produced and then used to produce other goods and services. Unlike consumer goods that satisfy wants directly, capital goods facilitate future production. Examples include manufacturing facilities, machinery, transportation equipment, and tools.
- (d) **Entrepreneurship** represents the creative effort of individuals who combine the other three factors of production to create goods and services. Entrepreneurs bear risk, organize production, and drive innovation.

Core Concept 1.2: Opportunity Cost

When making economic decisions, individuals and societies must choose among mutually exclusive alternatives. The **opportunity cost** of a choice represents the value of the best alternative forgone. This concept is fundamental to rational decision-making because choosing one option necessarily means sacrificing another.

Core Concept 1.3: Production Possibilities Curve

The **Production Possibilities Curve (PPC)** illustrates the maximum combinations of two goods that an economy can produce given its available resources and technology.

- (a) Points **on the curve** represent **productive efficiency** where all resources are fully utilized.
- (b) Points **inside the curve** indicate underallocation of resources
- (c) Points **outside the curve**—although appearing unattainable—can occur temporarily during periods of economic overheating when resources are used beyond their sustainable capacity. However, this state is not sustainable and typically leads to inflation, resource depletion, or economic crisis.

The **shape of the PPC** reveals important information about opportunity costs. A concave curve (*Figure 1*) indicates increasing opportunity costs, meaning that producing additional units of one good requires sacrificing increasingly larger quantities of the other good. This typically reflects the fact that resources are not equally adaptable to producing both goods.

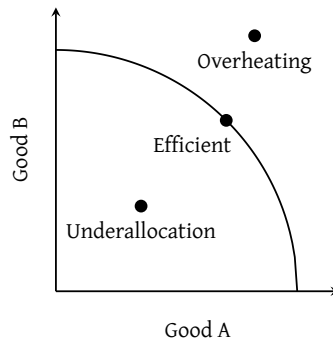


Figure 1: Production possibilities curve

Shifts in the PPC represent economic growth or decline. **Outward shifts indicate economic growth**, resulting from increases in available resources, improvements in technology, or enhancements in human capital. On the other hand, **inward shifts indicate economic decline**, potentially from natural disasters, war, or resource depletion.

Core Concept 1.4: Comparative and Absolute Advantage

An economy has an **absolute advantage** when it can produce more of a good than another economy using the same resources. **Comparative advantage** is when an economy can produce a good at a lower opportunity cost than another economy.

It is impossible for one economy to have a comparative advantage in both goods simultaneously. This principle, known as the **law of comparative advantage**, implies that all economies can benefit from trade by specializing in goods where they hold a comparative advantage. The gains from trade arise from this specialization and the subsequent exchange of goods.

Core Concept 1.5: Demand

Demand represents the quantity of a good that consumers are willing and able to purchase at various prices during a given time period. The **Law of Demand** states an inverse relationship between price and quantity demanded: as prices rise, quantity demanded falls, and vice versa. This relationship reflects both the **substitution effect** (consumers shift to cheaper alternatives) and the **income effect** (higher prices reduce purchasing power).

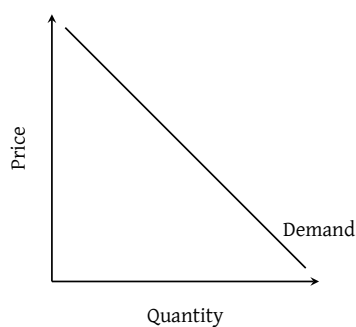


Figure 2: Demand curve

The determinants of demand, factors that can cause the entire demand curve to shift, include:

- (a) **Consumer income:** Normal goods experience increased demand as income rises, while inferior goods experience decreased demand.
- (b) **The number of buyers** in the market.
- (c) **The prices of substitute and complementary goods.**
- (d) **Consumer expectations** about future prices.
- (e) **Consumer tastes and preferences.**

Core Concept 1.6: Supply

Supply represents the quantity of a good that producers are willing and able to offer for sale at various prices. The **Law of Supply** indicates a direct relationship between price and quantity supplied: higher prices incentivize producers to increase production.

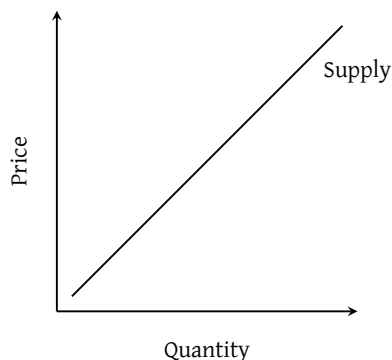


Figure 3: Supply curve

Determinants of supply include:

- (a) **Production costs**, including input prices and technology.
- (b) **The prices of related goods** firms could produce.
- (c) **Expectations about future prices.**
- (d) **The number of sellers** in the market.
- (e) **Government policies**, such as taxes, subsidies, and regulations.

Core Concept 1.7: Market Equilibrium

Market Equilibrium occurs where the supply and demand curves intersect. At this point, the quantity supplied equals the quantity demanded, and the market clears. The equilibrium price and quantity are determined automatically through the price mechanism without external intervention. Correspondingly, adjustments in equilibrium price occur when supply and demand curve shifts.

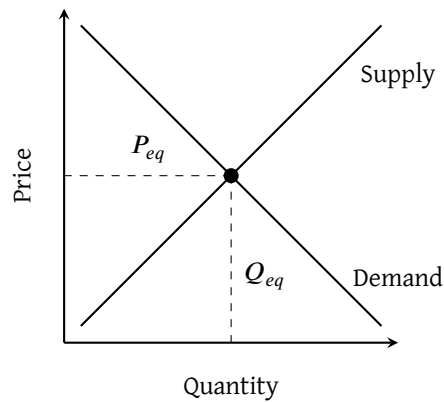


Figure 4: Market equilibrium

Core Concept 1.8: Circular Flow of Income

The **Circular Flow of Income** illustrates how goods, services, and money flow between **households** and **firms** in an economy. Firms produce goods and services using factors of production provided by households and, in exchange, pay households wages, rent, interest, and profits. Households then use this income to purchase goods and services from firms, completing the circular flow.

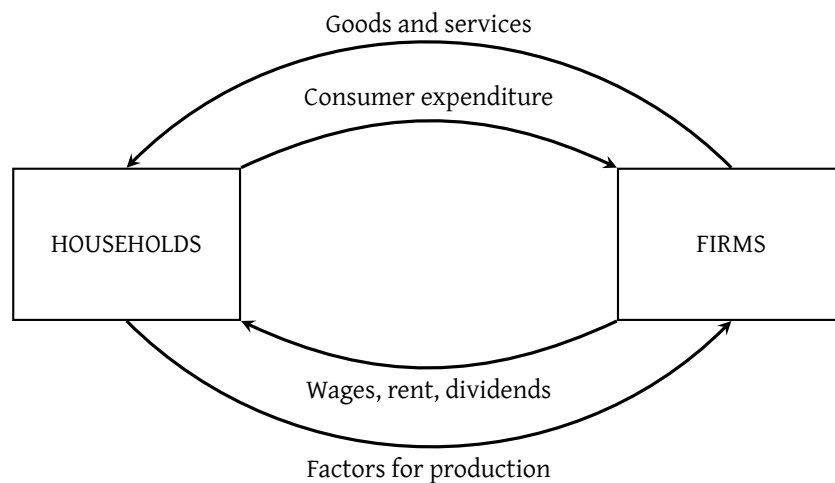


Figure 5: Circular flow of income; factor market (down) and product market (up)

This model (Figure 5) illustrates the interdependence between sectors of the economy and forms the basis for measuring aggregate economic activity through Gross Domestic Product (Core Concept 2.1).

2 Economic Indicators

Macroeconomic analysis relies on various **economic indicators** to assess the health and performance of an economy. These indicators provide insight into economic trends, business cycles, and the effectiveness of policy interventions.

2.1 Gross Domestic Product

Core Concept 2.1: Gross Domestic Product

Gross Domestic Product (GDP) represents the total market value of all **final** goods and services produced within a country's borders during a specific time period, typically one year or one quarter. GDP does not include the following items.

- (a) **Transfer payments:** Government payments not tied to current production, such as Social Security.
- (b) **Intermediate goods:** Which are counted in final goods to avoid double-counting.
- (c) **Financial transactions,** like stock purchases.
- (d) **Goods produced in previous periods,** such as existing home sales.

In general, GDP serves as the primary measure of economic output and standard of living.

Core Concept 2.2: Economic Cycles

The **business cycle** represents fluctuations in economic activity around the long-term growth trend. The cycle comprises four phases: expansion (*real GDP rising*), peak (*the highest point of economic activity*), recession (*real GDP falling*), and trough (*the lowest point*).

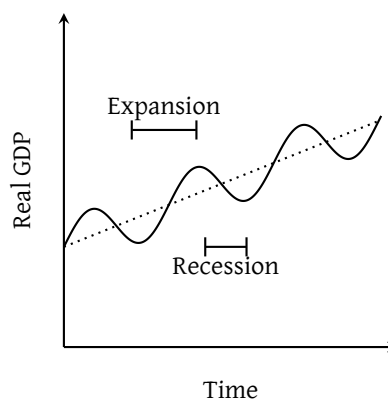


Figure 6: Business cycles

As shown in Figure 6, the **output gap** measures the difference between actual GDP and potential GDP (*the level of GDP that would exist if all resources were fully utilized*). A positive output gap indicates the economy is operating above its sustainable capacity, while a negative output gap indicates underutilization of resources.

Economists typically define a **recession** as two consecutive quarters of negative real GDP growth, though the National Bureau of Economic Research uses more comprehensive criteria.

GDP can be calculated through multiple approach that would yield equivalent results

Core Concept 2.3: Calculating GDP (Expenditures Approach)

The **expenditures approach**—the most common approach—sums all spending on final goods and services.

$$\text{GDP} = C + I + G + X - M$$

Here,

- (a) **C** represents household consumption (*spending on final goods and services*).
- (b) **I_g** represents gross investment (*spending on capital goods and inventory, excluding intermediate goods*).
- (c) **G** represents government purchases (*government spending on goods and services, excluding transfer payments*).
- (d) **X** represents exports.
- (e) **M** represents imports.

Core Concept 2.4: Net Domestic Product (NDP)

Net domestic product (NDP) is GDP but accounts for the depreciation of capital goods. Its calculation takes the same method as the expenditures approach (Core Concept 2.3), but substitutes gross investment with net investment. **Net investment (I_n)** is gross investment minus depreciation (*the decrease in value of capital goods over time*).

While GDP is the headline figure mainstream media usually follows, NDP is a more honest figure because it measures only the investment used for additional growth on the existing capital base, excluding the portion that goes into simply replacing worn down capital.

Core Concept 2.5: Calculating GDP (Income Approach)

The **income approach** calculates GDP by summing all incomes earned in production. This includes wages and salaries, interest income, rental income, profits (*including proprietor income and corporate profits*), and statistical adjustments (*depreciation, sales taxes, and net foreign factor income*). The equality of expenditure and income approaches reflects the circular flow of income.

$$\text{GDP} = \text{Wages} + \text{Interest} + \text{Rental Income} + \text{Profits} + \text{Statistical Adjustments}$$

Core Concept 2.6: Calculating GDP (Value Added Approach)

The **value added approach** calculates GDP by summing the value added at each stage of production. Value added equals the value of a firm's output minus the value of intermediate goods it purchases. This approach avoids double-counting by measuring only the incremental value contributed at each production stage.

$$\text{GDP} = \sum \text{Value Added at Each Stage}$$

For example, consider bread production: a farmer sells wheat for \$1, a mill sells flour for \$2, and a bakery sells bread for \$3. The total value added is $\$1 + (\$2 - \$1) + (\$3 - \$2) = \3 , which equals the final selling price.

The three methods of calculating GDP—expenditure, income, and value-added—are essentially three ways of counting the same dollar as it travels through the economy's circular flow. The expenditure approach catches that dollar at the cash register when a final good is purchased, the income approach tracks it as it is paid out to workers and business owners, and the value-added approach measures it as it is "built" piece-by-piece at every stage of production. Because every transaction is a closed loop where one person's spending is another person's income and a direct reflection of the new value created, all three paths inevitably lead to the exact same total: the market value of everything produced in a year—or GDP.

Core Concept 2.7: Limitations of GDP

Despite its widespread use, GDP has important limitations as a measure of economic well-being.

- (a) GDP excludes **non-market transactions** and **underground economic activity**, such as household production, volunteer work, and illegal transactions.
- (b) GDP does not capture changes in **environmental quality**, **income inequality**, or **social conditions**. A country with high GDP may also have significant pollution, income disparity, or social unrest.
- (c) GDP does not account for **depreciation**, which overstates true economic welfare. NDP, as outlined in Core Concept 2.4, accounts for this inaccuracy.

Core Concept 2.8: Economic Growth

Economic Growth refers to a sustained increase in real GDP per capita (*per person*) over time, representing the expansion of an economy's productive capacity and living standards.

Long-run economic growth is represented by rightward shifts in the long-run aggregate supply curve (*Core Concept 3.4*) and outward shifts in the PPC (*Core Concept 1.3*). The aggregate production function expresses total output as a function of capital stock and human capital.

$$Y = A \cdot f(L, K, H, N)$$

Here,

- (a) *Y*: Total Real Output (Real GDP).
- (b) *A*: Total Factor Productivity (TFP). This represents technology, efficiency, and the "know-how" of the economy.
- (c) *L*: Labor. The total number of workers or hours worked.
- (d) *K*: Physical Capital. Tools, machinery, factories, and infrastructure.
- (e) *H*: Human Capital. The skills, education, and experience of the workforce.
- (f) *N*: Natural Resources. Land, minerals, and raw materials.

Labor productivity (*output per worker*) is a key determinant of economic growth. Factors driving growth include:

- (a) **Capital accumulation**: Investment in machinery and infrastructure
- (b) **Human capital development**: Education and skills training
- (c) **Technological advancement**
- (d) **Increases in the labor force participation rate** (*Core Concept 2.9*)

2.2 Labor Market

The **labor market** provides crucial information about employment, unemployment, and the economy's productive capacity.

Core Concept 2.9: Labor Force Participation Rate

The **labor force participation rate** (LFPR) measures the proportion of the working-age population actively in the labor market. It is calculated by the following equation.

$$\text{LFPR} = \left(\frac{\text{Labor Force}}{\text{Civilian Non-institutional Population}} \right) \times 100$$

The **labor force** is defined as the sum of all individuals who are either currently working or actively seeking work, excluding discouraged workers who have stopped seeking employment. The **civilian noninstitutional population** represents the "pool" individuals who are legally and physically able to participate in the economy.

Core Concept 2.10: Unemployment Rate (U-3)

The **unemployment rate** (or also called the *U-3 rate*) measures the proportion of unemployed persons to the total labor force. This is a major headline statistic that all news outlets, economists, investors, etc. follow.

$$\text{Unemployment Rate} = \left(\frac{\text{Number of Unemployed}}{\text{Labor Force}} \right) \times 100$$

For many advanced economies, including the U.S., and unemployment rate of 3.5% to 4.5% is considered the "sweet spot" that governments strive for.

Core Concept 2.11: Types of Unemployment

Unemployment can be further categorized into the following baskets.

- (a) **Frictional unemployment** occurs when workers are between jobs or searching for new employment. This type of unemployment is inevitable and even desirable in a dynamic economy, as it reflects workers seeking better matches.
- (b) **Structural unemployment** arises from mismatches between worker skills and job requirements, often due to technological change or shifts in the industrial composition of the economy. This unemployment persists even when the economy is at full employment.
- (c) **Cyclical unemployment** results from economic recessions and fluctuations in aggregate demand. When the economy contracts, employers lay off workers, increasing cyclical unemployment.

The **natural rate of unemployment (NRU)** equals the sum of frictional and structural unemployment. It represents the unemployment rate when the economy is at full employment, consistent with stable inflation.

Core Concept 2.12: Underemployment (U-6)

The **underemployment rate** (also known as the *U-6 rate*) represents the proportion of unemployed and underemployed persons to the total labor force. Underemployed persons consists of **marginally attached workers** (people who want a job and have looked for one in the last 12 months, but not in the last 4 weeks) and **part-time workers for economic reasons** (people who want full-time work (35+ hours) but are working part-time because their hours were cut or they cannot find a full-time position).

$$\text{Underemployment Rate} = \left(\frac{\text{Unemployed} + \text{Underemployed}}{\text{Labor Force} + \text{Marginally Attached}} \right) \times 100$$

Many economists argue that the U-6 rate provides a much more accurate picture of "labor underutilization" because it includes people who are technically working but are struggling.

Core Concept 2.13: Other Indicators

The following are other important labor market indicators.

- (a) **Nonfarm Payrolls:** The change in the number of employed workers excluding farm workers, private household employees, and nonprofit employees. This is the primary gauge of labor market strength.
- (b) **Quits Rate:** The number of employees who voluntarily left their jobs during the reporting period, divided by total employment. A higher quits rate indicates worker confidence in finding better opportunities, signaling a strong labor market.
- (c) **Jobs Opening Rate:** The number of job openings at the end of the reporting period, divided by the sum of employment and job openings. This measures employer demand for labor and is a leading indicator of future hiring and wage growth.
- (d) **Average Hourly Earnings:** The average gross hourly wages paid to nonfarm private employees, excluding government workers. Changes in this measure signal wage inflation pressures and are watched closely as a component of labor costs.
- (e) **Initial Claims:** The number of new unemployment insurance claims filed by recently unemployed workers. Lower initial claims indicate a stronger labor market, while rising claims may signal economic weakening.

2.3 Inflation

Core Concept 2.14: Inflation

Inflation represents a sustained increase in the general price level, equivalent to a decrease in the purchasing power of money.

Inflation is primarily kept track of by the following two statistics.

Core Concept 2.15: Price Indices

The **Consumer Price Index (CPI)** is an inflationary measure which tracks the cost of a market basket of goods and services purchased by a typical consumer. This figure is a highly tracked statistic by new outlets and investors.

$$\text{CPI} = \frac{\text{Cost of market basket in current prices}}{\text{Cost of market basket in base period prices}} \times 100$$

The **Producer Price Index (PPI)** measures the average change over time in selling prices received by domestic producers for their output. Unlike the CPI, which tracks prices from the consumer perspective, the PPI reflects inflation from the producer's viewpoint. The PPI serves as a **leading indicator** of consumer inflation because cost changes at the producer level are eventually passed downstream to consumers. While less prominent in popular media than the CPI, the PPI remains a closely monitored inflation gauge among economists and policymakers.

Both the CPI and PPI have **core** versions—**core CPI** and **core PPI**—which exclude food and energy prices from their calculations. These components are omitted because they exhibit high volatility driven by natural factors (such as weather and crop yields) and geopolitical events (such as oil supply disruptions). By excluding these volatile elements, core measures provide smoother, more stable trends compared to their raw counterparts.

For the U.S., these figures are maintained by the Bureau of Labor Statistics (BLS) and are the main inflation figures used by the U.S. government for administrative and legal purposes.

Core Concept 2.16: Personal Consumption Expenditures

The **Personal Consumption Expenditures (PCE)** index is another inflation index that captures a broader range of consumer spending and accounts for substitution effects. This inflation measure is also another highly tracked statistic.

Like price indices, the PCE index also has a core version for the same reasons.

For the U.S. this figure is maintained by Bureau of Economic Analysis and is the preferred inflation gauge for the Federal Reserve in the context of setting the Federal Funds Rate (Core Concept 5.7).

Core Concept 2.17: Inflation Rate

The **inflation rate** equals the percentage change in the price level:

$$\text{Inflation Rate} = \frac{\text{Current Inflation Figure} - \text{Previous Inflation Figure}}{\text{Previous Inflation Figure}} \times 100$$

Economists and news outlets typically prioritize the **Year-over-Year (YoY)** rate because it provides a more stable picture of long-term trends by filtering out "noise" and seasonal fluctuations that often distort the **Month-over-Month (MoM)** data.

$$\text{MoM \%} = \left(\frac{\text{Value}_{\text{currentmonth}} - \text{Value}_{\text{previousmonth}}}{\text{Value}_{\text{previousmonth}}} \right) \times 100$$

$$\text{YoY \%} = \left(\frac{\text{Value}_{\text{currentmonth}} - \text{Value}_{\text{samemonthlastyear}}}{\text{Value}_{\text{samemonthlastyear}}} \right) \times 100$$

Core Concept 2.18: Nominal versus Real Variables

Nominal variables are measured in current dollars and are not adjusted for inflation. Examples include **nominal GDP** (*measured in current prices*) and nominal interest rates (*the stated rate without inflation adjustment*).

Real variables are adjusted for inflation and reflect actual purchasing power. **Real GDP** uses constant base-year prices. For instance, the following is the equation for **real interest rate**.

$$\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$$

However, most real interest rates reflect the expected future shift in inflation rather than realized inflation, thus the more accurate relationship is:

$$\text{Real Interest Rate} \approx \text{Nominal Interest Rate} - \text{Expected Inflation}$$

This distinction is crucial for economic decision-making, as borrowers and lenders care about real costs and returns. Lenders prefer to increase their lending rates when inflation is expected to rise, while borrowers prefer to delay borrowing when inflation is expected to fall. Consequently, inflation expectations serve as

a focal point in determining the quantitative terms of debt contracts and financial arrangements. Many contracts, including loans, bonds, and wage agreements, are written in nominal terms, meaning their real value depends heavily on the inflation rate over their duration.

Unexpected inflation redistributes wealth from creditors to debtors. If a lender sets a nominal interest rate expecting low inflation but actual inflation turns out to be high, the real return becomes negative, benefiting the borrower at the lender's expense. This uncertainty complicates financial planning and investment decisions.

The **GDP deflator** provides another measure of the price level, representing how much of nominal GDP growth is just due to inflation.

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

2.4 Housing

Housing indicators provide valuable insights into the broader economic outlook. These statistics are closely watched because of housing's unique role in the economy.

Core Concept 2.19: Housing and Its Effects on the Economy

Although housing represents only approximately 3% of the economy, it serves as a powerful economic indicator for several reasons. Housing purchases typically involve 30-year mortgages, making housing finance deeply intertwined with the financial system. Additionally, housing purchases trigger purchases of complementary goods, including furniture, appliances, and home improvement materials.

The strong correlation between housing activity and GDP makes housing indicators valuable for predicting economic trends. The 2008 financial crisis demonstrated how housing market troubles can cascade through the entire financial system, particularly through mortgage-backed securities and derivatives tied to residential real estate.

Core Concept 2.20: Housing Indicators

The following are the major housing indicators followed by economists

- (a) **Housing Starts:** The number of new residential construction projects that have begun during the reporting period. This is one of the first indicators to decline at the onset of a recession and one of the last to recover, making it a reliable leading indicator of economic trends.
- (b) **New Housing Sales:** The number of newly constructed homes with a signed sales contract. New home sales represent a smaller portion of the market than existing sales but are considered a leading indicator because they reflect consumer confidence in the economy and housing market.
- (c) **Existing Housing Sales:** The number of previously constructed homes with a completed transaction. Existing home sales make up the majority of housing market activity and reflect the supply side of the market, including homeowner decisions to sell.
- (d) **Mortgage Rates:** The interest rates charged on home loans, most commonly tracked for 30-year and 15-year fixed-rate mortgages. Mortgage rates directly affect housing affordability and demand, with lower rates typically stimulating home purchases and higher rates slowing the market.

2.5 Sentiment Indicators

Core Concept 2.21: Economic Sentiment

Economic Sentiment is a composite indicator that measures the overall "mood" or optimism of an economy. It is a **leading indicator**, meaning it often predicts changes in GDP before they occur.

Sentiment is typically subdivided into two primary pillars:

- **Business Confidence:** Measures the expectations of firm managers regarding production, order books, and inventory levels. It predicts future Investment (I).
- **Consumer Confidence:** Measures households' plans for major purchases and their outlook on personal finances. It predicts future Consumption (C).

Core Concept 2.22: Key Consumer Sentiment Indicators

University of Michigan Consumer Sentiment: A monthly survey of at least 500 households that measures consumers' attitudes toward current and expected financial conditions, including their assessment of present economic conditions and their expectations for the future. Readings greater than 50 indicate positive consumer sentiment and typically signal stronger future consumer spending.

Conference Board Consumer Confidence: A monthly survey of approximately 3,000 households that assesses consumer attitudes toward business conditions, employment, and income. The index comprises two components: the present situation index (*current economic conditions*) and the expectations index (*short-term outlook for business conditions, employment, and income*).

University of Michigan Inflation Expectations: A survey-based measure of consumers' expectations for inflation over the next one to five years. These expectations directly influence wage demands, pricing decisions, and actual inflation outcomes, making this a closely watched indicator for monetary policy.

Core Concept 2.23: Key Business Sentiment Indicators

ISM Manufacturing PMI: A monthly survey of purchasing managers at over 300 manufacturing firms conducted by the Institute for Supply Management. The PMI summarizes whether business conditions have improved or worsened. A reading above 50 indicates expansion, while below 50 indicates contraction.

NFIB Small Business Optimism: A monthly survey of small business owners conducted by the National Federation of Independent Business that measures their expectations for the economy, hiring plans, capital expenditure plans, and earnings.

2.6 Economic Calendar and Indicator Releases

Core Concept 2.24: World Economic Calendar

The **world economic calendar** is a comprehensive schedule of economic indicator releases (such as GDP, CPI, and employment data) and central bank meetings throughout the year.

It is the primary tool used by investors and policymakers to track **market expectations** versus **actual results**. Discrepancies between the "forecast" and the "actual" data typically drive volatility in global currency and stock markets.

Category	Indicator	
GDP	Nominal GDP	Nominal GDP per capita
	Real GDP	Real GDP per capita
Inflation	CPI	PPI
	PCE	
Labor Market	U-3 Rate	U-6 Rate
	Participation Rate	Nonfarm Payrolls
	Quits Rate	Jobs Opening Rate
	Average Hourly Earnings	Initial Claims
Housing	Housing Starts	New Housing Sales
	Existing Housing Sales	30-Year Mortgage Rate
	15-Year Mortgage Rate	
Sentiment	ISM Manufacturing PMI	UMich Consumer Sentiment
	CB Consumer Confidence	UMich Inflation Expectations
	NFIB Small Business Optimism	
Other	VIX	Dollar Index
	Federal Funds Rate	

Table 1: Key economic indicators by category

3 Macroeconomic Relationships

3.1 Aggregate Demand

Core Concept 3.1: Aggregate Demand

Aggregate Demand (AD) represents the total quantity of goods and services demanded in the economy at different price levels.

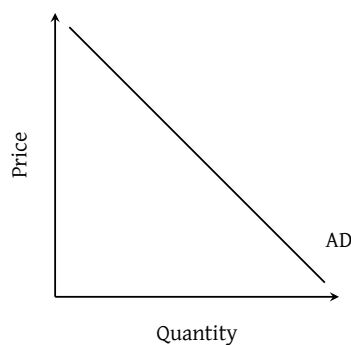


Figure 7: Aggregate demand curve

The AD curve slopes downward for three reasons:

- (a) **Wealth effect:** Lower price levels increase real wealth, allowing consumers to purchase more goods and services.

When prices at the store drop, the \$20 bill in your pocket can suddenly buy more stuff. You haven't received a raise, but you feel richer because your money has more "purchasing power" and subsequently increase consumer spending (C).

- (b) **Interest rate effect:** Lower price levels reduce money demand, causing interest rates to fall, stimulating investment.

When prices are low, you don't need to carry as much cash for your daily coffee and groceries. You leave that extra money in the bank. When banks have more money sitting in accounts, they "on sale" their loans by lowering interest rates to get people to borrow, raising investment and thus increasing investment (I).

- (c) **Exchange rate effect:** Lower interest rates cause currency depreciation, making domestic goods cheaper abroad and stimulating exports.

In conjunction with the interest rate effect, when interest rates in the U.S. drop, foreign investors also move their money elsewhere to find better returns. To do that, they sell their U.S. Dollars. This makes the Dollar "weaker" or cheaper. Suddenly, American products look like they are "on clearance" to people in other countries.

Rightwards Shifts in the AD curve result from increases in the following four components that are not due to a change in price. Correspondingly, decreases shift the AD leftward.

- (a) **Consumer spending:** Increases from tax cuts (*through the tax multiplier effect; Core Concept 3.2*) or increased confidence shift AD rightward.
- (b) **Investment:** Increases from business optimism or technological innovation shift AD rightward.
- (c) **Government spending:** Increases shift AD rightward. *Government spending acts as the "balancer" to manually shove the AD curve back to the LRAS (Core Concept 5.1) when the private sector fails to do so.*
- (d) **Net exports:** Increases from foreign economic growth or currency depreciation (*note that depreciation at face value is a price change from the foreign buyer perspective, but the domestic price is unchanged, making this a non-price influence*) shift AD rightward.

Core Concept 3.2: The Multiplier Effect

An initial change in spending creates a larger final change in GDP through the multiplier effect. To understand why this is, we introduce two new concepts: the **Marginal Propensity to Consume or MPC** (*the fraction of additional disposable income that households spend*) and the **Marginal Propensity to Save (MPS)** (*the fraction of additional household income that households save*). Thus,

$$MPC + MPS = 1$$

The **spending multiplier** effect operates through successive rounds of spending. When one person spends income, the recipient receives income and spends a portion (*the MPC*), continuing until leakages (*saving, taxes, imports*) absorb the remaining income.

$$\text{Spending Multiplier} = \frac{1}{1 - MPC} = \frac{1}{MPS} = 1 + MPC + MPC^2 + MPC^3 + \dots$$

As a result, a government spending stimulus of \$X would increase aggregate demand by $\frac{\$X}{MPS}$.

The **tax multiplier** works in the same manner but with a smaller initial impact because a tax change first affects disposable income rather than direct aggregate demand. When taxes are cut, consumers do not spend the entire amount; they save a portion (*the MPS*) immediately. Consequently, the first round of the spending

cycle is "diluted" by this initial leakage, making the tax multiplier's overall effect less powerful than that of the spending multiplier. The negative sign in the following equation for the tax multiplier reflects that tax increases reduce disposable income and spending.

$$\text{Tax Multiplier} = -\frac{\text{MPC}}{1 - \text{MPC}} = -\frac{\text{MPC}}{\text{MPS}} = \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots$$

Consequently, a total tax change of $-\$X$ would increase aggregate demand by $-\frac{\$X \cdot \text{MPC}}{\text{MPS}}$.

3.2 Aggregate Supply

Core Concept 3.3: Short-Run Aggregate Supply

Short-Run Aggregate Supply (SRAS) represents the total output firms are willing to produce at different price levels in the short run, when **some input costs (particularly wages) are fixed**.

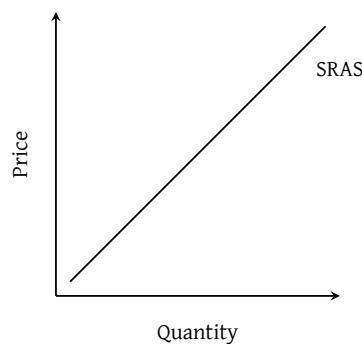


Figure 8: Short-run aggregate supply

The SRAS curve slopes upward because higher price levels increase profitability (*higher margins*) when input costs are **sticky**. Firms respond to higher prices by increasing production to capture the temporary opportunity for additional profits until production costs adjust, representing a movement along the SRAS curve (*Core Concept 3.5*).

Shifts in the SRAS result from changes in the following.

- (a) **Productivity:** More productive factors of production (*technological advancement, machinery, etc.*) shift the SRAS rightward.
- (b) **Input Costs:** Lower input costs (*such as decreasing oil prices*) mean producers have larger margins, shifting the SRAS rightward.
- (c) **Labor Market Conditions:** High unemployment among the labor force pushes wages downward because firms have more bargaining power over its employees. Wages are a type of input costs, thus decreases in wages (*stemming from labor market conditions*) will shift the supply curve rightward.
- (d) **Taxes/Subsidies:** Governments can incentivize or disincentivize the production of goods and services by selectively applying subsidies (*artificially raising margins*) or additional taxes (*artificially lowering margins*) on firms, respectively.

Core Concept 3.4: Long-Run Aggregate Supply

Long-Run Aggregate Supply (LRAS) represents the economy's potential output when all input costs can fully adjust to price changes (*the reality in the long run*). Here, output is independent of the price level and is equivalent to potential GDP.

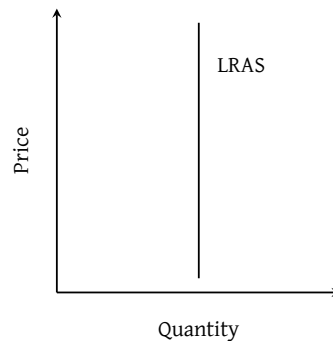


Figure 9: Long-run aggregate supply

3.3 Economic Theory

Core Concept 3.5: Shifts in Short Run Equilibrium

The short run equilibrium price and quantity can be shifted by any shift in the AD and SRAS curve, thus we can classify these shifts into four categories.

- (a) **AD curve shifts rightward.** This scenario occurs when consumer demand and business growth occurs at a faster rate at which the economy can sustainably handle in the short term, leading to **high inflation** (increase in P) and **low unemployment** (increase in Q). This scenario is also called **demand-pull inflation** and is essentially an "economic boom" (high GDP growth).

*It's important to point out here the relationship between the quantity of goods and services denoted in the AD-SRAS-LRAS plot and GDP. Since GDP is essentially the market value of all products and services produced, it is in a direct relationship with Q . In order to support the production of more goods and services, an economy needs more workers and thus unemployment would subsequently fall. This observation is also in line with **Okun's Law**, which is the observed statistical relationship between a country's GDP and its unemployment rate.*

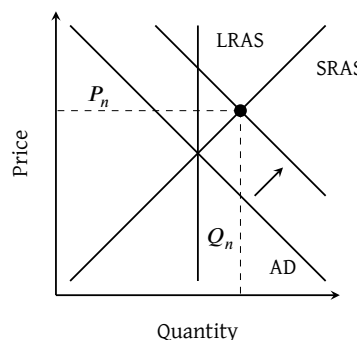


Figure 10: AD shifts rightward; associated with unsustainable economic growth

- (b) **AD curve shifts leftward.** This scenario occurs when consumer demand and business growth slows down

(consumer pessimism, higher interest rates, tax hikes to reduce government deficit, etc.). Consequently, this leads to **low inflation** (decrease in P) and **high unemployment** (decrease in Q). This scenario represents a recession (low GDP growth).

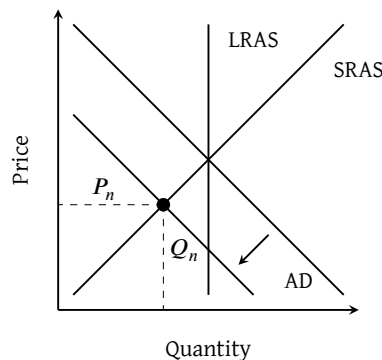


Figure 11: AD shifts leftward; associated with a recession

- (c) **SRAS curve shifts rightward.** This scenario is the consequence of a positive supply shock, an unexpected event that suddenly changes the supply of a product or commodity (sustained lower oil prices, infrastructure development), leading to **low inflation** (decrease in P) and **low unemployment** (increase in Q). This is the "golden scenario" that economies strive for because allows for lower prices and higher economic growth simultaneously.

Positive supply shocks are usually common with technological breakthroughs and lower input costs. However, although this may seem like an ideal scenario, there are some **negative consequences of this societal development**.

Structural displacement of workers due to automation is a large downside (such as AI displacing the current white collar workforce). Lower prices can induce a **deflationary spiral risk** during which AD shifts left because consumers want to wait for prices to drop further. Since debt erodes in value under high inflationary conditions, **borrowers undertake a comparatively higher debt burden** as a positive supply shock may cause wages to also fall.

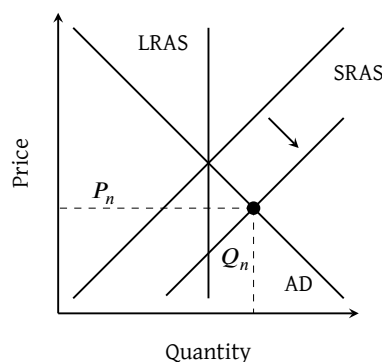


Figure 12: SRAS shifts rightward; associated with sudden positive supply shocks

- (d) **SRAS curve shifts leftward.** This scenario is the consequence of a negative supply shock, an unexpected event that suddenly changes the supply of a product or commodity (such as the oil shock of 1973 which led to the Volcker shock), leading to **high inflation** (increase in P) and **high unemployment** (decrease in Q). This scenario is also called **cost-push inflation**.

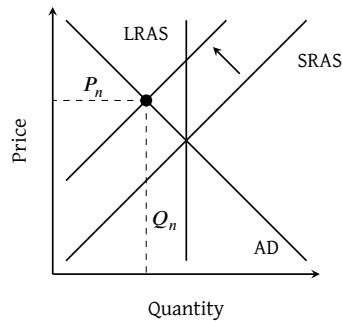


Figure 13: SRAS shifts leftward; associated with sudden negative supply shocks

Core Concept 3.6: Long-Run Self-Adjustment & Equilibrium

Without government intervention, the economy naturally adjusts to long-run equilibrium through wage and price flexibility.

- (a) **Rightward AD Shift Correction:** Because unemployment is low, workers have high bargaining power, leading to **higher nominal wages** to keep up with the rising cost of living. Consequently, SRAS shifts to the left to account for the higher input cost and the economy returns to LRAS at a **higher price level**.

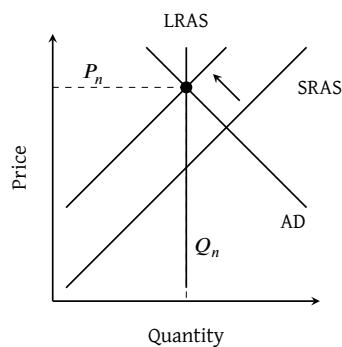


Figure 14: Rightward AD shift correction; SRAS shifts left

- (b) **Leftward AD Shift Correction:** Because unemployment is high, workers have low bargaining power, which forces them to accept **lower nominal wages**. As a result, SRAS shifts to the right to account for lower input costs and the economy returns to LRAS at a **lower price level**.

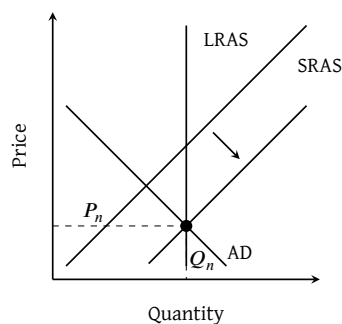


Figure 15: Leftward AD shift correction; SRAS shifts right

- (c) **Rightward SRAS Shift Correction/Integration:** The correction for this scenario depends on the time frame of the positive supply shock in question.

If **temporary**, the SRAS simply shifts back to the right because higher employment will increase wages and thus input costs, decreasing output eventually.

If **permanent**, the supply shock is usually due a positive economic breakthrough, thus an economy would integrate the productivity gain rather than "correct" it. Consequently, LRAS will shift to the right and sustainably decrease the NRU of that economy. *This is why developed nations tend to have a lower unemployment rate than developing nations.*

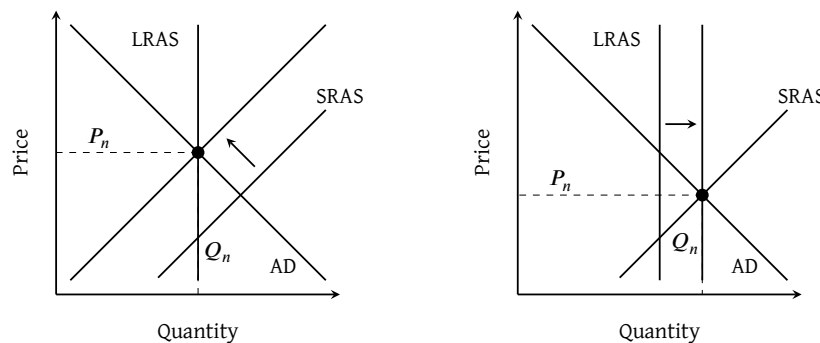


Figure 16: Rightward SRAS shift correction/integration; SRAS shifts back left if temporary (left) or LRAS shifts right under permanent positive supply shocks (right)

- (d) **Leftward SRAS Shift Correction:** The nature of the correction for this scenario also depends on the time frame of the supply shock in question.

If **temporary**, the SRAS simply shifts back to the right once the "input costs" return to normal.

If **permanent**, workers eventually accept the new reality of higher prices and lower output. High unemployment consequently pushes wages downwards (due to lower worker bargaining power) and shifts SRAS back right due to this lowered input cost.

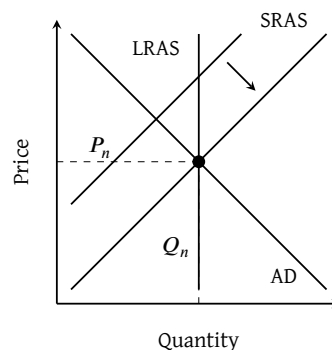


Figure 17: Leftward SRAS shift correction; SRAS shifts back

The general self-adjustment process explains why classical economists emphasized the economy's natural tendency toward full employment (U-3 Rate = NRU), though the adjustment process may be slow and painful.

Core Concept 3.7: The Phillips Curve

The **Philips Curve** is a graphical representation of the relationship between unemployment and inflation.

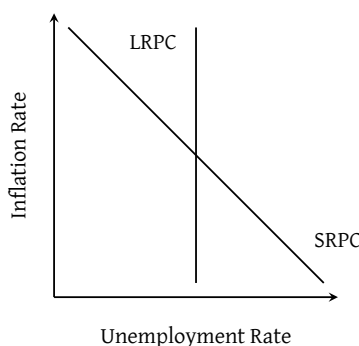


Figure 18: Short-run and long-run Phillips curve

In the short run, unemployment and inflation share an inverse relationship. Importantly, movements along and entire shifts of the SRPC directly corresponds to shifts in the AD and SRAS curve, respectively.

In the long run, the economy returns to the natural rate of unemployment regardless of inflation, making the LRPC vertical.

- (a) **Movements along the SRPC** occur when AD shifts (*equivalent to a movement along the SRAS*), leading to an inverse change in both inflation and unemployment.

Keep in mind here that the subsequent SRAS shift that occurs to correct the AD curve shift leads the SRPC to shift horizontally (not shown) so that the unemployment rate sits at the NRU.

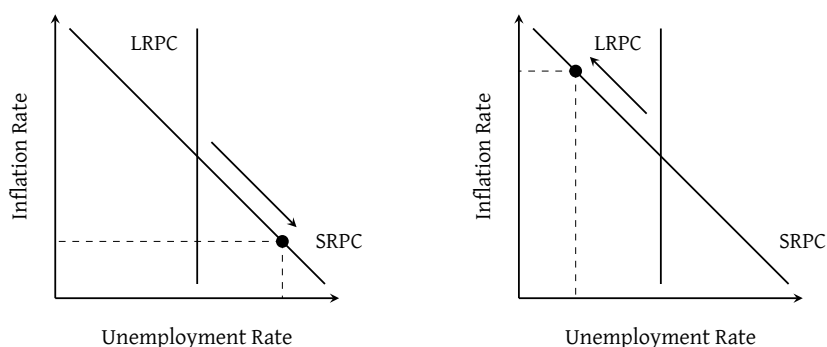


Figure 19: Movements along the SRPC when AD shifts left (*left; demand decreases*) and right (*right; demand increases*)

- (b) **Shifts in the SRPC** occur when SRAS shifts (*positive or negative supply shocks*), leading to the same directional change in both inflation and unemployment.

Under negative supply shocks, SRAS eventually shifts back to the long run equilibrium, equivalent to the SRPC shifting back to its initial position. Under positive supply shocks, either the SRPC shifts back to its initial position (if the shift is temporary) and or the LRPC shifts to the left (if the shift is permanent) and consequently leads to a sustainably lower NRU. This equivalent correction on the SRPC-LRPC plot is not shown.

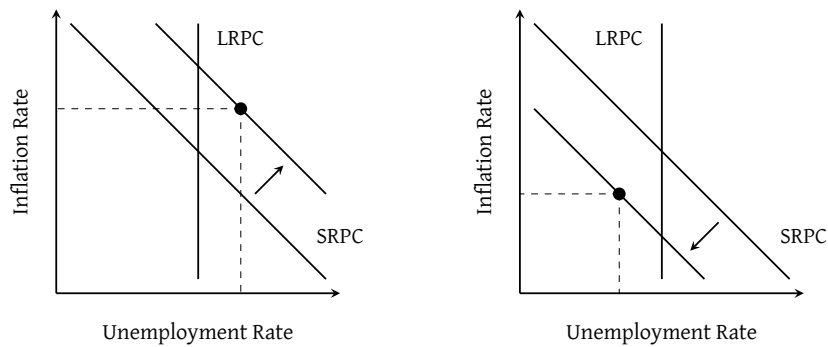


Figure 20: Shifts in the SRPC when SRAS shifts left (left; negative supply shock) and right (right; positive supply shock)

Core Concept 3.8: Quantity Theory of Money

The **Quantity Theory of Money** provides a mathematical framework for understanding the long-run causes of inflation. The theory is expressed through the following identity.

$$MV = PQ$$

Here, M represents the **money supply**, V represents **velocity** (*how frequently a unit of money changes hands*), P represents the **price level**, and Q represents the **quantity of real output** (*or real GDP*).

Classical economists assume that both velocity (V) and real output (Q) are relatively stable or determined by non-monetary factors in the long run.

- **Velocity** is determined by spending habits and banking technology.
- **Output** is determined by the economy's productive capacity (*labor, capital, and technology*).

Because V and Q are assumed to be stable, any change in the money supply (M) must result in a proportional change in the price level (P). Therefore, sustained inflation—a persistent rise in P—is interpreted as the direct result of a sustained increase in the money supply.

4 Financial Sector

The financial sector channels savings to investment and facilitates economic transactions.

4.1 Money and the Money Market

Core Concept 4.1: Functions of Money

Money serves the following three essential functions in an economy.

- **Medium of exchange:** Money facilitates transactions by acting as an intermediary instrument and eliminating barter's double coincidence of wants (*we two traders have exactly what each other want*).
- **Store of value:** Money preserves purchasing power for future use well compared to other assets (*unlike food, for instance, because it rots*).
- **Unit of account:** Money provides a common measure for valuing goods and services across a country.

Note that money is not a perfect store of value or unit of account due to inflation. Overtime, its purchasing power does indeed

erode. During hyperinflation, its unit of account also may skew based on the varying sentiment of market participants.

The **opportunity cost of holding money** is the interest that could be earned by holding interest-bearing assets instead (such as savings accounts or treasury bills). Higher interest rate conditions increase this opportunity cost, reducing money demand because people would much rather earn interest from securities.

Core Concept 4.2: Money Supply (M0, M1, and M2)

Economists categorize the money supply using the following aggregates.

- **M0:** Physical currency in circulation and reserves held by banks at the central bank.
- **M1:** M0 money supply, checking accounts, and travelers' checks.
- **M2:** M0/M1 money supply, savings deposits, small time deposits (CDs), and money market mutual funds.

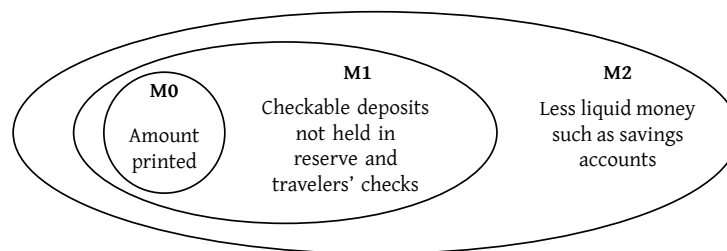


Figure 21: Diagram of money supply

These aggregates reflect different degrees of **liquidity**, with broader measures (M2) capturing more forms of money that cannot be easily converted to cash compared to the foundational counterparts (M0 or M1).

Core Concept 4.3: The Money Market

The money market determines the equilibrium nominal interest rate through the interaction of money supply and money demand. The **money supply curve is vertical** because it is a fixed value that is manipulated by the government and central bank through fiscal policy (Core Concept 5.1) and monetary policy (Core Concept 5.4).

On the other hand, **money demand slopes downward** because higher interest rates increase the opportunity cost of holding money, thus market participants would rather deposit their money into interest-bearing assets.

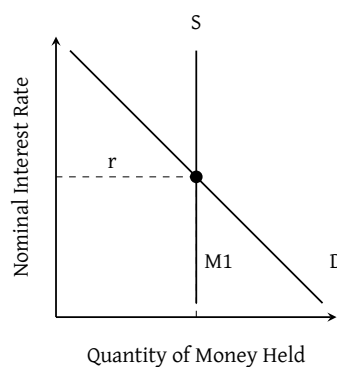


Figure 22: Money market

The following non-interest-rate factors lead to shifts in the money demand curve.

- (a) **Price Level:** When the overall price of goods and services increases, individuals must hold more money to execute these transactions, shifting the money demand curve rightward.
- (b) **National Income:** When people earn more (*national income rise*), they spend more, raising the demand for money and shifting the curve rightward.
- (c) **Technology Affecting Liquidity:** When technology makes it easier/faster to convert assets into "spendable" forms, the need for holding liquid cash lowers, shifting the money demand curve leftward.

4.2 Banking System

Core Concept 4.4: Fractional Reserve Banking

Banks operate under **fractional reserve banking**, maintaining only a fraction of deposits as **reserves** while lending the remainder to outside market participants. This reserve to lending ratio is set by the **required reserve ratio (RRR)**, which establishes the minimum fraction of deposits that banks must hold as reserves.

This system creates money through the multiplier effect as shown by the following formulas where $rr = \text{reserve ratio} \geq \text{RRR}$.

$$\text{money multiplier} = \frac{1}{rr} = 1 + rr + rr^2 + rr^3 + \dots$$

As a result, the total change in the money supply of an economy for a given amount of excess reserve (*money a bank does not need to store as required reserves*) is shown by the following equation.

$$\Delta \text{Money Supply} = \frac{\text{excess reserves}}{rr}$$

Note that excess reserves, in layman's terms, refers to an injection of money from the government or central bank (new money). It's important to notice that this is not talking about consumer deposits, which are already accounted for in the money supply and is simply recirculated money.

As these loans are spent and redeposited into other banks, the process repeats across the banking system, mathematically expanding a single initial deposit into a much larger total increase in the money supply. We have seen this same pattern with the multiplier effect in the context of MPC and MPS (Core Concept 3.2). With a 20% RRR, an initial deposit of \$1,000 can ultimately support \$5,000 in total deposits.

Core Concept 4.5: The Reserve Market

The reserve market describes the supply and demand for reserves within the banking system at a given **policy rate**—the benchmark interest rate set by the central bank (*for the U.S., this is the federal funds rate, covered in Core Concept 5.7*).

As shown in Figure 23, the **supply curve is vertical** because the quantity of reserves is determined unilaterally by the central bank through monetary policy (Subsection 5.2). The **demand curve**, by contrast, consists of three sections: an upper ceiling, a downward sloping middle region, and a lower floor.

- The **ceiling** is set by the **discount rate** (Core Concept 5.7) and represents the maximum interest rate any bank in the economy is willing to pay for reserves.
- The **floor** is set by the **interest on reserves (IOR) rate** (Core Concept 5.7) and represents the minimum interest rate any bank is willing to accept.

- The **downward sloping region** between them represents the range over which market interest rates can adjust in response to shifts in reserve supply.

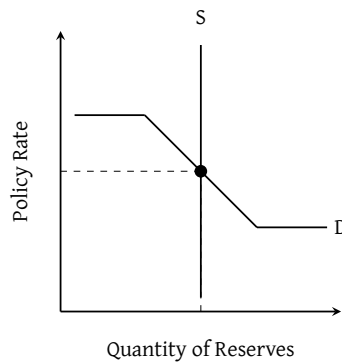


Figure 23: Reserve market plot

The reserve market can further be characterised by two regimes depending on the position of reserve supply.

- (a) **Limited reserves:** Reserves are scarce, so banks can charge higher interest rates to borrowers who have few alternatives. This corresponds to the region to the left of the floor in the reserve market diagram.

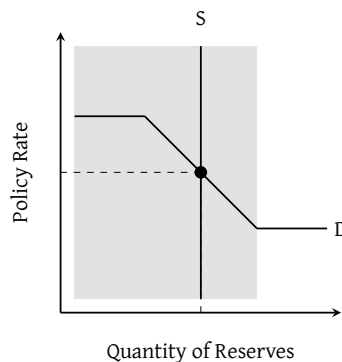


Figure 24: Limited reserves section of the reserve market plot

- (b) **Ample reserves:** Reserves are abundant, so banks must lower their rates to attract borrowers. This corresponds to the region to the right of the floor and reflects an increase in the money supply.

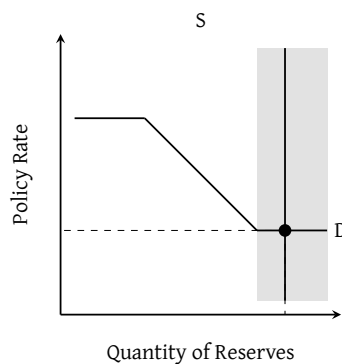


Figure 25: Ample reserves section of the reserve market plot

The prevailing reserve regime also determines which tools are available to the central bank.

- When the policy rate lies **between the floor and ceiling**, the central bank can influence it directly by adjusting the **required reserve ratio** (Core Concept 5.6) or conducting **open market operations** (Core Concept 5.5).
- When the policy rate is **pinned at the ceiling or floor**, the central bank can only shift it by adjusting the **discount rate** or the **IOR rate**, respectively. *This policy implementation results in a corresponding vertical elongation or compression of the demand curve.*

Core Concept 4.6: Loanable Funds Supply and Demand

The loanable funds market determines the real interest rate through the supply of funds (savings provided by lenders) and demand for funds (investments sought after by borrowers).

Real interest rates represent the price of borrowing from the supply of loans available, analogous to price on a simple supply and demand plot.

Thus when real interest rates rise, the quantity of loanable funds demanded decreases (movement up the demand curve), and when real interest rates fall, the demand subsequently increases because loans have gotten "cheaper" (movement down the demand curve). Supply curve sloping upwards works in the same manner but in the context of supply.

The position of these curves reflects national saving behavior (supply), government borrowing (demand), and investment profitability (demand) expectations.

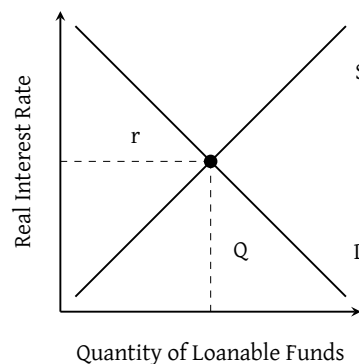


Figure 26: Loanable funds

Core Concept 4.7: National Savings

National Savings represents total saving in an economy for investment, equaling private savings plus public savings.

A **closed economy** national savings is represented by the following equation.

$$I = \text{National Savings} = Y - C - G = \underbrace{(Y - T - C)}_{\text{Private Saving}} + \underbrace{(T - G)}_{\text{Public Saving}}$$

As the above equations shows nicely, private saving is simply household income (Y) after taxes (T) and private consumption (C) and public saving (or the government saving) its tax revenue (T) after accounting for its spending (G).

The national saving in a closed economy context is equal to investment (I) and is an alternative representation of the supply of loanable funds (*savings*) and demand for those loanable funds (*investment*) at equilibrium.

In an **open economy**, the relationship between savings and investment changes because a country is no longer limited to its own internal resources. It can interact with the rest of the world and must consider **net capital outflow (NCO)**.

$$\text{National Savings} = I + \text{NCO}$$

Net Capital Outflow (NCO) measures the net flow of funds being **invested abroad**, calculated by the following equation.

$$\text{NCO} = \text{Domestic purchase of foreign assets} - \text{Foreign purchase of domestic assets}$$

5 Stabilization Policies

Without government intervention, the economy can potentially spiral out of control, characterized by extreme high and lows of the natural business cycle due to fluctuating aggregate demand levels. In order to stabilize this naturally volatile economic timeline, policymakers use fiscal and monetary to manage aggregate demand.

5.1 Fiscal Policy

Core Concept 5.1: Fiscal Policy

Fiscal policy involves **government spending** and **taxation decisions** to influence aggregate demand.

- **Contractionary fiscal policy** reduces AD to combat inflation by decreasing government spending ($\downarrow G$) and increasing taxes ($\downarrow C$).
- **Expansionary fiscal policy** stimulates AD to combat recession by increasing government spending ($\uparrow G$) and decreasing taxes ($\uparrow C$).

*Keep in mind that the initial impact of fiscal policy is multiplied through the **spending and tax multiplier** (Core Concept 3.2), meaning the ultimate change in GDP exceeds the initial fiscal impulse of government spending or tax policy.

Core Concept 5.2: Automatic Stabilizers

Automatic stabilizers are fiscal policies that automatically counteract economic fluctuations **without discretionary action by the government**. These stabilizers provide immediate response to economic conditions, avoiding recognition and implementation lags that affect discretionary policy.

Progressive income taxes exemplify automatic stabilizers.

- During **economic expansions**, rising incomes push taxpayers into higher brackets, automatically increasing tax revenues and exerting downward pressure on the economy.
- During **economic recessions**, falling incomes reduce tax burdens, providing automatic stimulus.

Core Concept 5.3: Crowding Out and Government Debt

The **crowding out effect** occurs when government spending reduces private sector activity, potentially nullifying the original intent of stimulating the economy in times of recession. This is particularly problematic

when the government is operating at a **deficit** (*annual spending greater than tax revenue*), consequently building its **debt** (*accumulated deficits*).

Governments generally finance this spending through two primary channels.

- (a) **Loans** (*Core Concept 5.5*) are usually the sole option for developed nations under a deficit. However, when an institution as large as a government enters the loanable funds market, the demand curve of loanable funds shifts rightward, increasing real interest rates, and consequently **crowding out investment (I)**.

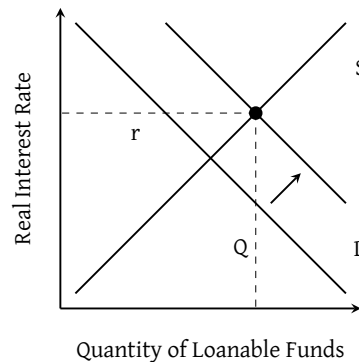


Figure 27: Government induced loanable funds demand increase

- (b) **Tax hikes** are the second option, although less common because nations that operate under perpetual debt use it primarily as a tool for preventing debt from compounding uncontrollably. However, tax hikes also directly **slow down consumer spending (C) and investment (I)**, which is also called **tax-induced contraction**.

5.2 Monetary Policy

Core Concept 5.4: Monetary Policy

Monetary policy involves central bank actions controlling the **money supply** and **interest rates**.

- **Expansionary monetary policy** stimulates economy activity to combat recession by decreasing interest rates and increasing the total money supply.
- **Contractionary monetary policy** reduces economy activity to combat inflation by increasing interest rates and decreasing the total money supply.

The central bank specifically influences rates through the following monetary policy tools.

Core Concept 5.5: Monetary Policy Tool (*Open Market Operations*)

Open market operations involve buying and selling government bonds to affect private sector bank reserves and thus the total money supply of the economy.

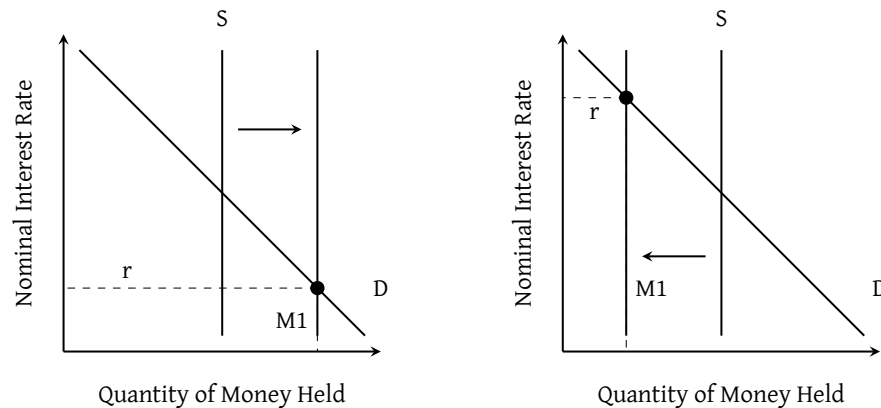


Figure 28: Money supply increasing (left) and decreasing (right) as a result of government purchasing and selling bonds, respectively

- **Purchasing bonds** involves the government handing over money to the private sector in exchange for its bonds, thus expanding the money supply and consequently lowering interest rates in order to combat recession.
- **Selling bonds** involves the government receiving money from the private sector in exchange for its bonds, thus contracting the money supply and consequently raising interest rates in order to combat inflation.

Core Concept 5.6: Monetary Policy Tool (*Required Reserve Ratio*)

Changes in the **required reserve ratio (RRR)** affect banks' lending capacity. Lowering the RRR expands the money supply and raising it contracts the money supply in the same manner as in Core Concept 5.5.

Note that the ultimate change in money supply is also amplified due to the multiplier effect in fractional reserve banking (Core Concept 4.4).

Historically, the RRR was set by the Federal Reserve and publicly announced. However, as of March 2020 (COVID pandemic), the Fed reduced the RRR to 0% for all deposits, effectively eliminating it as an active policy tool.

In contrast, some central banks (such as the People's Bank of China) continue to actively adjust reserve requirements as a monetary policy tool.

Core Concept 5.7: Monetary Policy Tool (*Discount Rate, Interest on Reserves, and Federal Funds Rate*)

The **Federal Funds Rate** is the target interest rate range set by the central bank in order to broadly manage economic activity. This range of this rate is controlled by the discount rate (*ceiling*) and the interest on reserves rate (*floor*).

- The **discount rate** is the interest rate banks pay to borrow from the central bank.
Consequently, no bank would ever rationally borrow from another commercial bank with a higher interest rate than the discount rate. Thus, lowering the discount rate encourages borrowing because it effectively acts as the ceiling for all other interest rates; raising it has the opposite effect.
- The **interest on reserves rate (IOR)** is the rate of interest that a central bank pays to commercial banks on the funds they hold in their reserve accounts at the central bank.
This effectively sets a floor for interest rates because no lender would ever rationally lend at a rate below the IOR.



Figure 29: Federal funds rate (July 1954 - February 2026)

The FFR serves as the primary benchmark for the global financial system, effectively anchoring the interest rates for all other types of borrowing. To maintain economic stability, the **Federal Open Market Committee (FOMC)** meets eight times annually to evaluate and adjust the FFR target.

The Federal Reserve operates under a dual mandate, seeking to maintain an economy characterized by price stability—quantified as a 2% long-run inflation target—and maximum employment, which is traditionally associated with a NRU centered around 4%.

6 International Trade

6.1 Balance of Payments

The **Balance of Payments (BOP)**—composed of the **current account** and the **capital/financial account**—is a comprehensive record of all economic transactions between the residents of a country and the rest of the world over a specific period.

Core Concept 6.1: Current Account

The **current account** measures the net balance of **value** flow into a country, composed of the following three components.

- (a) **Balance of Trade:** The balance of trade encompasses the value of **exports** (*money coming in*) minus **imports** (*money going out*), including both physical goods and services.

$$\text{Balance of Trade} = \text{Exports} - \text{Imports}$$

- (b) **Primary Income:** Primary income refers to net earnings on investments and wages from business entities. This is recorded as a positive when recipients are domestic entities (*citizens and business of that country*) receiving earnings (*dividends, interest, etc.*) from foreign investments or wages from foreign businesses. On the other hand, negative primary income refers to when recipients are foreign entities (*citizens and businesses of a foreign country*) receiving earnings from domestic investment and wages from domestic businesses.

$$\text{Primary Income} = \text{Earnings}_{\text{domestic}} + \text{Wages}_{\text{domestic}} - (\text{Earnings}_{\text{foreign}} + \text{Wages}_{\text{foreign}})$$

- (c) **Secondary Income:** Secondary income—often called **current transfers**—refers to money or goods

transferred without anything given in return. Note that in balance of trade, goods and services were given in return for money, and in primary income, either labor or capital was given in return for earnings or wages. Secondary income transfers are usually one of the following.

- **Personal transfers** occurs when a person living in one country send money directly to a person in another country.
- **Institutional/Government Transfers (Foreign Aid)** refers to money or resources given by one government to another. This is like personal transfers but government entities as the sender and recipient.
- **Taxes and Social Contributions** tracks money crossing borders for social welfare purposes. For example, a citizen of Country A works remotely for a business entity at Country B. If Country B taxes that worker's income, the tax (taken as a portion of the original income outflow) is recorded as inflow for Country B's Secondary Income and an outflow for Country A

An outflow of value would coorespond to a negative transfer in the current account, whereas an inflow would be recorded as a positive transfer.

$$\text{Secondary Income} = \text{Inflow} - \text{Outflow}$$

Core Concept 6.2: Capital Account

The **capital account** is a record of "capital transfers" and the acquisition/disposal of **non-produced, non-financial assets**, specifically referring to the ownership transfer of this that does not involve a traditional financial instruments like a loan or stock.

The capital account behaves similarly to the current account in the context of representing value transfers, however the transfered "object" in question is usually not tangible, whereas, say, a transfer of cash for secondary income is still obtainable in paper form.

- **Debt Forgiveness:** When a government forgives the debt of another, the "value" of that forgiven debt is recorded.
- **Intangible Assets:** The sale or purchase of things that aren't "produced" by labor, such as patents, copyrights, trademarks, and franchises.

Core Concept 6.3: Financial Account

The **financial account** tracks the ownership of financial assets between domestic and foreign entities. It reflects **how a country finances the value transfers in its current and capital accounts**. A transfer of a financial asset from a foreign to domestic entity (*import*) would be considered a negative line item. On the other hand, a transfer of a financial asset from a domestic to foriegn entity (*export*) would be considered a positive line item.

Keep in mind that cash is considered a financial asset.

Core Concept 6.4: Balancing the Current, Captial, and Financial Accounts

In theory, the absolute value of the current account (CA) and capital account (KA) should equal to that of the financial account (FA).

$$|CA + KA| = |FA| \quad \text{or} \quad CA + KA = -FA$$

*In reality, accurately keeping track of all cross border transactions that occur for a country is impossible. Thus official government data usually includes a line item called **Net Errors and Omissions** to force a balance.*

6.2 Exchange Rates

Core Concept 6.5: Exchange Rates

The **exchange rate** represents the price of one currency in terms of another. Currency **appreciation** means a currency becomes more valuable relative to another, and **depreciation** means it becomes less valuable.

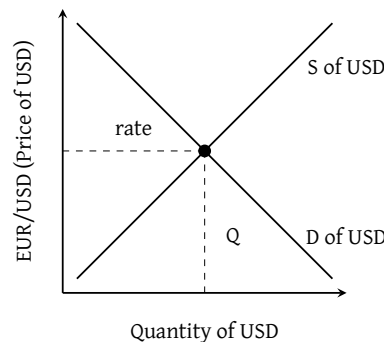


Figure 30: USD/EUR currency market

Like other markets, exchange rates are determined by supply and demand. Factors affecting currency supply and demand include the following.

- **Relative interest rates:** Higher rates attract investors towards assets that are bought via that currency.
- **Relative price levels:** Lower prices for a country's goods and services increase global economy competitiveness and create more demand for the relevant currency due to higher purchasing power.
- **Relative national income:** Higher income increases the domestic currency supply, increasing relative demand for all goods, including imports, which decreases demand for the domestic currency.
- **Tariffs:** Tariffs reduce import demand for foreign goods and services, preventing domestic currency demand from falling.

Core Concept 6.6: Real Interest Rate Parity

The **Real Interest Rate Parity** condition states that differences in real interest rates between countries drive capital flows. Higher domestic real interest rates relative to foreign rates attract capital inflows, appreciating the domestic currency. These **capital flows continue until real interest rates equalize** across countries, implying that in integrated capital markets, real interest rates theoretically should converge globally.

In reality, real interest rates do not perfectly converge and serves rather as a long-run tendency than a constant state of dynamic equilibrium due to the following.

- **Transaction Costs:** Fees and regulations that impede the free capital flow between countries.
- **Risk Premiums:** Higher rates in certain economies are required to compensate for political or default risk in specific countries.
- **Non-Tradable Goods:** Items like rent or local services (*like a haircut*) cannot be exported to take advantage

of for price differences, preventing local inflation rates from equalizing, which in turn prevents real interest rates from converging globally.